ABHINIT SATI

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RESEARCH INTERESTS

- Inverse Design and Physics-based Optimization for Manufacturing
- Computational Fabrication, Geometry Processing and Physics-based Simulation

EDUCATION

- PhD, Computer Science, Boston University, 2021–present (expected June 2027)
- MS, Computer Science, Clemson University, 2019–2021, GPA 3.7/4.0
- BS, Computer Science, University of Mumbai, 2016–2019, CGPA 8.2/10.0

RESEARCH EXPERIENCE

Graduate Research Assistant — **Boston University**
Advised by Dr. Emily Whiting; co-advised by Dr. Edward Chien — *September 2021 to present*

- **Fluidic shaping.** Developed a segmentation algorithm to produce sparse 3D printed frames; such that when liquid polymer is injected into the submerged frame, interfacial tension shapes it into the approximate target geometry upon curing. Structures come out with $\approx 1\ \mathrm{nm}$ surface smoothness (in collaboration with Dr. Valeri Frumkin's group at BU).
- **Fabric formwork.** Developed a physics aware mesh segmentation method for designing sewn fabric molds. The mold pieces are computed such that local membrane stretching under loading of plaster/concrete aligns with fiber and seam directions. This reduces twisting and shearing of the fabric and produces casts closer to the target geometry.
- **Topology-optimized reinforced concrete.** Used the fabric formwork method to build low-cost, topology-optimized reinforced concrete beams (in collaboration with Dr. Josephine Carstensen's group at MIT).

Graduate Research Assistant — **Clemson University**
Advised by Dr. Victor Zordan and Dr. Ioannis Karamouzas — *January 2020 to May 2021*

- **Inverse Design.** Developed algorithms to simulate and predict cloth behavior given initial loading conditions. Implemented an inverse-design optimization scheme to efficiently find designs that produce desired performance.
- **Planning algorithms.** Developed stitch-planning techniques to alter the tensile properties of fabric, including path planning for directional stitch control that produces anisotropic material behavior within the limits of embroidery machines.

PUBLICATIONS

Peer-reviewed

- **Abhinit Sati**, Tiffany Bao, Jeff Tedi, Edward Chien, Emily Whiting. "Strain-Field Based Segmentation for Fabric Formwork." *Computer Graphics Forum* (Proc. Eurographics 2026), 45(2), 2026. doi:10.1111/cgf.70367
- Mark Hellrich, **Abhinit Sati**, Jackson L. Jewett, Edward Chien, Emily Whiting, Josephine V. Carstensen. "Affordable Construction of Topology Optimized Reinforced Concrete Through Fabric Formwork." Journal of Building Engineering, 2026. 10.1016/j.jobe.2026.117472
- **Abhinit Sati**, Ioannis Karamouzas, Victor Zordan. "DIGISEW: Anisotropic Stitching for Variable Stretch in Textiles." *Symposium on Computational Fabrication (SCF)*, 2021. doi:10.1145/3485114.3485121

Preprints

- **Abhinit Sati**, Jacob Garrett, Valeri Frumkin, Edward Chien, Emily Whiting. "Piecewise CMC Segmentation for Fluidic Shaping." arXiv:2611.00227, 2026. arxiv.org/pdf/2611.00227
- Jacob Garrett, **Abhinit Sati**, Edward Chien, Emily Whiting, Valeri Frumkin. "Manufacturing Complex Structures Through Interfacial Self-Organization." Preprint forthcoming, arXiv, 2026.

Master's Thesis

- **Abhinit Sati**. *A System for Programming Anisotropic Physical Behaviour in Cloth Fabric.* MS thesis, Clemson University, 2021. tigerprints.clemson.edu/all_theses/3621

INVITED TALKS

- School of Computing Seminar, Clemson University.

AWARDS

- Graduate Scholarship, Clemson University: fully funded Research Assistantship for the final academic year.
- Runner-up, Best Virtual Reality Project (course project; instructor Dr. Sabarish Babu): an immersive 3D application for visualizing molecular orientation and bond formation to aid learning of chemistry students.